

Emily Tumacder

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EDUCATION

Purdue University

Aug 2023 - May 2027

B.S Computer Engineering, Minor in Statistics, Honors Goss Scholar

- GPA: 3.76/4.00
- *Specialization:* Computer Systems

EXPERIENCE

Hardware Engineering Intern

May 2026 - Aug 2026

Stryker-Vocera, San Jose CA

- Prototyped visual and haptic feedback components for Vocera SyncBadge
- Experimented with BLE beaconing compatibility on SyncBadge and created a demo implementation for hospital use
- Developed hardware and programmed (Python) a GUI for power consumption data collection used in validation of PCB function

Electrical Engineering Intern

May 2025 - May 2026

Kimberly Clark, Paris TX

- Designed electrical packages for upgrading machinery and IT systems, including PLC programs, safety circuits, and routing
- Programmed a PyGame-based training simulation for line operators, focused on scalability using multithreading and OOP principles
- Used Splunk (SPL), VSphere, and OTBase to troubleshoot 100+ network switch statuses during capital projects and built documentation for plant wide network devices

Digital Systems Teaching Assistant

Jan 2026 - May 2026

Purdue University, West Lafayette IN

- Facilitated three lab sections with 30+ students each
- Debugged SystemVerilog code during design labs
- Diagnosed wiring issues in student breadboard circuits during hardware labs

RESEARCH & PROJECTS

Smart Ear Cuff

Present

Senior Design

- Building an ear-worn device for biometric data collection, including posture and heart rate

End-to-End Aphasia Severity Prediction

Jan 2024 - Dec 2025

Advised by Dr. Yan Cong, CLA @ Purdue University

<https://doi.org/10.3389/frai.2026.1857106>

- Demonstrated the use of LLMs in automatic aphasia detection on clinical conversations through the finetuning and prompt engineering of a Llama3.2 model
- Extracted and filtered examples of speech from a database of 50k+ annotated lines of patient conversational data using Python (pandas)

BiBi: Benchtop Buddy

Dec 2025

ECE362: Intro to Microelectronics

- Designed the hardware for an autonomous desktop cleaning robot using KiCAD and simulations via SPICE

SKILLS

Programming

Python, SystemVerilog, C, HTML/CSS, Java

Engineering Tools

AutoCAD, KiCAD, ControlLogix, TINA-TI SPICE, DMM, Oscilloscope, Spectrum analyzer, Function generator

Statistical Tools

R, MiniTab, Splunk, MATLAB, SQL